

ORIGINAL ARTICLE

A GARCH-temporal fusion transformer model for the volatility prediction of exchange traded funds

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Abstract

Forecasting the volatility of financial assets—defined as the degree of their price variation over time—is a trending topic in financial research. Enhancing prediction accuracy is crucial in this field, as an asset's volatility is widely used to assess the risk associated with its returns. In this context, we introduce a novel hybrid model that integrates traditional econometric techniques, specifically GARCH models, with the Temporal Fusion Transformer (TFT), a cutting-edge deep learning architecture. We designed such a model for analyzing Exchange Traded Funds (ETFs) composed of assets from the S&P 500, a benchmark index tracking 500 large US companies therefore reflecting the overall health and trends of the stock market, across various sectors. Utilizing volatility proxies such as historical volatility (HV) and the Garman–Klass (GK) method, our study demonstrates that the hybrid GARCH-TFT model significantly outperforms alternative models in forecasting the GK proxy and achieves performance comparable to the stand-alone TFT model for HV, underscoring the potential of merging machine learning approaches with traditional econometric methods to enhance predictive precision in volatile financial markets.

Keywords Volatility · Exchange traded funds · ETF forecasting · Econometrics · Deep learning · GARCH · Temporal fusion transformer

1 Introduction

In managing financial assets, striking a strategic balance between risk and return is essential for enhancing investment value in line with an investor's risk tolerance. Achieving this balance requires a thorough analysis of the volatility of investment returns, a critical aspect of risk management [1, 2]. Volatility measures the degree of variation in a financial asset's price over time, capturing both the frequency and magnitude of price fluctuations to indicate the risk associated with the asset's returns.

Econometrics has been fundamental in analyzing volatility, leading to models like Autoregressive Conditional Heteroskedasticity (ARCH) and its generalized version (GARCH) [3]. Moreover, advances in computational capabilities have enabled the development of innovative time series analysis techniques, including deep learning models. In the last decade, a synergy has emerged between traditional econometric models and advanced computational frameworks, particularly involving multilayer perceptron (MLP), gated recurrent units (GRU), and long short-term memory (LSTM) networks. These models are often used in hybrid approaches alongside established methods to achieve more accurate volatility predictions [4].



However, the research in the artificial intelligence field, particularly in deep learning, keeps moving forward at a wide pace. In 2017, the introduction of the Transformer architecture [5] marked a major advancement in natural language processing, significantly influencing the analysis of textual data. Recently, the unique features of such an architecture have been adapted for time series analysis, leading to the development of models like the Temporal Fusion Transformer (TFT) [6], or other more specific solutions as space-specific graph convolutional recurrent transformer [7].

As highlighted in [8], there is still a large gap between modern machine learning models and those applied to volatility forecasting, moreover, the integration of traditional econometric models like GARCH with TFT, holding promises of enhancing volatility forecasting accuracy, without dismissing the possibility of having an explicable model due to the gated attention mechanisms integrated in the TFT. Managers and investors benefit from more accurate volatility forecasts by making better-informed decisions, optimizing investment strategies, and improving risk management practices. Policymakers also gain from precise volatility predictions, which are crucial for evaluating and crafting regulatory interventions. With a clearer understanding of market dynamics, they can implement measures to stabilize financial markets and promote economic stability, safeguarding against systemic risks and ensuring a resilient economic environment.

Contributions In this context, this work addresses the research question: *How can the integration of the TFT with traditional GARCH models enhance the accuracy of volatility forecasts?* To answer this question, we evaluated our approach on a portfolio of Exchange Traded Funds (ETFs) reflective of major companies listed on the Standard & Poor's 500 (S&P 500 index, which includes 500 leading companies and covers approximately 80% of available market capitalization).

By introducing this novel hybrid model that combines the strengths of both traditional econometric methods and advanced deep learning architectures, we make the following contributions:

- To the best of our knowledge, this article introduces the first architecture that directly combines GARCH and TFT within a single unified framework. Unlike existing hybrid GARCH approaches that primarily combine econometric models with simpler machine learning techniques, our model leverages the recurrent architecture and attention mechanisms of TFT to achieve superior forecast performance. This hybrid framework has proven to be a significant advancement over both traditional approaches and cutting-edge hybrid or non-hybrid models.
- We provide a comprehensive review of the features used to train the model, detailing their hierarchical importance in enhancing predictive accuracy. This not only underscores the robustness of our approach but also offers deeper insights into how these features contribute to the improved performance, setting our methodology apart within the existing literature.

In particular, readers could find the repository of the work at <https://github.com/lbacco/garch-tft>.

2 Related works

In econometric modeling, significant focus is often placed on deterministic models, which define variables of interest as functions of predetermined factors. This section specifically examines the role of (G)ARCH models within this framework, emphasizing their application in financial variable modeling [9–11], including their use with high-frequency data [12]. (G)ARCH models are founded on the concepts of heteroscedasticity and conditional volatility, recognizing that the volatility of asset returns is dynamic and varies over time in response to specific conditions or events. These models effectively capture the changing risk associated with financial assets by modeling how volatility evolves as depicted in [13].

After exploring the fundamentals and applications of (G)ARCH models, the section introduces the deep learning approaches to volatility forecasting that are increasing in number in the last years as shown in [14]. Then, it focuses on discussing hybrid methods that integrate (G)ARCH models with artificial neural networks (ANNs).

These hybrid approaches aim to leverage the strengths of both traditional econometric models and advanced machine learning techniques to enhance the accuracy and efficiency of volatility predictions. By combining the structural advantages of (G)ARCH models with the pattern recognition capabilities of ANNs, these hybrid models offer improved performance in forecasting tasks, addressing the limitations inherent in using either approach alone.

2.1 (G)ARCH models

2.1.1 ARCH

In 1982, G. Engle introduced the ARCH (p) model [15] with the intention of modeling conditional volatility through an autoregressive approach. This model posits that future volatility can be predicted based on past volatility, operating under the assumption that fluctuations in financial data are often correlated with their preceding values. The model is mathematically defined by the following equations:

$$y_t = \mu_t + \sigma_t \eta_t \sim N(0, 1) \quad (1)$$

$$\varepsilon_t = \sigma_t \eta_t, \varepsilon_t | \chi_{t-1} \sim N(0, \sigma_t^2) \quad (2)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-p}^2 \quad (3)$$

Equation 1 includes μ_t , which can be predicted from the mean equation for the time series y_t , along with the error term ε_t , modeled as stochastic component $\sigma_t \eta_t$ where η_t is the standardized error term and σ_t is the standard deviation. Equation 2 illustrates that the error term follows a Gaussian distribution, given the condition that all information up to time $t - 1$ is known. Finally, Eq. 3 introduces the conditional variance at time as a function of past squared residuals. Additionally, to ensure that this variable remains positive, all estimated parameters must be nonnegative.

2.1.2 GARCH

One of the limitations of the ARCH model is its frequent requirement for numerous parameters and a high order of p to adequately capture the volatility process. In contrast, the GARCH (p,q) model, proposed by Bollerslev in 1986 [16], is based on a specific form of an infinite ARCH. This model allows for a reduction in the number of estimated parameters by imposing nonlinear constraints. This enhancement is facilitated by the introduction of the parameter q in Eq. 3, which incorporates past conditional variances. Consequently, this addition enhances the model's flexibility and adaptability.

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-p}^2 + \beta_1 \sigma_{t-1}^2 + \cdots + \beta_p \sigma_{t-q}^2 \quad (4)$$

In this approach, volatility is predicted as a weighted average of the previously predicted variance and the observed variance.

Both ARCH and GARCH models account for volatility clustering and leptokurtosis; however, their conditional variance depends solely on the magnitude, not the sign, of the underlying activity. This limitation hinders their effectiveness in accurately describing the “leverage effect.” To overcome this, various nonlinear extensions of GARCH models have been proposed. These include exponential, Glosten–Jagannathan–Runkle, and threshold GARCH, respectively, E-GARCH [17], GJR-GARCH [18], and T-GARCH [19].

2.1.3 E-GARCH

In Nelson's E-GARCH model, the asymmetry exhibited by volatility in response to positive and negative information is explicitly addressed. This is achieved by defining the conditional variance as a function of both the sign and the magnitude of such responses:

$$\log(\sigma_t^2) = \alpha_0 + \sum_{i=1}^p \alpha_i g(z_{t-i}) + \sum_{j=1}^q \beta_j \log(\sigma_{t-j}^2) \quad (5)$$

$$g(z_t) = \begin{cases} (\psi + \gamma)z_t - \gamma\mathbb{E}(|z_t|) & \text{if } z_t \geq 0 \\ (\psi - \gamma)z_t - \gamma\mathbb{E}(|z_t|) & \text{if } z_t < 0 \end{cases} \quad (6)$$

In particular, $g(z_t)$ turns out to be a linear combination of the sign z_t , and the modulus $|z_t|$, with the negative shock attributed to $(\psi - \gamma)$ and the positive on to $(\psi + \gamma)$. In the case of a negative shock associated with a negative value of z_t , the effect is obviously amplified. An advantage of using this model is the absence of constraints on the parameters, as the equation is defined on $\log(\sigma_t^2)$, thereby automatically satisfying the condition for a positive variance.

2.1.4 GJR-GARCH

A second methodology to incorporating asymmetric effects is proposed in Glosten et al. (1993) [18] through the GJR-GARCH model. In this model, the aforementioned effect is modeled using an additional term I_{t-1} .

$$\log(\sigma_t^2) = \alpha_0 + \sum_{i=1}^p (\alpha_i + \gamma_i I_{t-1}) \epsilon_{t-i}^2 + \sum_{j=1}^q \beta_j \log(\sigma_{t-j}^2) \quad (7)$$

where $\alpha_0 > 0$, $\alpha_i \geq 0$, $\alpha_i + \gamma_i \geq 0$, $\beta_j \geq 0$ for $i = 1, \dots, p$ and $j = 1, \dots, q$.

$$I_{t-1} = \begin{cases} 0 & \text{if } r_{t-1} \geq \mu \\ 1 & \text{if } r_{t-1} < \mu \end{cases} \quad (8)$$

Specifically, the model recognizes that the impacts of positive and negative returns on volatility can differ, thereby giving rise to the well-known asymmetries. In this context, these asymmetries are addressed by the additional term introduced in the model.

2.1.5 T-GARCH

The T-GARCH model (that is, the threshold GARCH model), is proposed by Zakoian in 1994 [19], with the previously introduced GJR-GARCH model. It introduces a distinct approach, focusing on structural or regime changes in financial data. This approach particularly emphasizes the observation of a threshold crossing, usually set at zero, by the lagged innovation. The model can be formalized as follows:

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q (\alpha_i + \varphi_i D_{t-i}) \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2 \quad (9)$$

$$D_{t-1} = \begin{cases} 0 & \text{if } \epsilon_{t-1}^2 \leq 0 \\ 1 & \text{if } \epsilon_{t-1}^2 > 0 \end{cases} \quad (10)$$

In this model, the variable D_{t-1} enables the model to respond to shocks based on the sign of the considered residual terms. Specifically, if a residual term is negative, D_{t-1} will be unitary, and the impact of negative shocks will be measured as the $\alpha_i + \varphi_i$ where the coefficient φ_i quantifies this effect. The expected sign of φ is positive, implying that volatility increases in response to negative innovations, and this increase is more pronounced as the size of the innovation grows. Finally, if $\varphi \geq 0$, the impact of news is asymmetric, affirming the existence of the leverage effect.

2.2 Deep learning approaches

Recent advancements in deep learning have influenced the development of purely neural network-based models for analyzing financial market volatility. These models capitalize on the powerful pattern recognition capabilities of deep learning architectures, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), to predict market behaviors without relying on traditional econometric models [20].

For instance, Chen and Huang [21] introduced a CNN-based model specifically for stock market volatility prediction. Their model adeptly captures dependencies in financial time series data, resulting in enhanced forecasting accuracy. Their study demonstrated that CNNs could effectively discern complex patterns in stock market data, outperforming traditional GARCH models in terms of mean absolute error and mean squared error.

Similarly, Ju [22] developed an LSTM model to forecast financial market volatility, with a particular focus on the SSE index. The study highlighted that the LSTM model provided superior forecasts compared to traditional time series models, although it noted the need for improvement in the accuracy of peak data predictions. This underscores the robustness of deep learning methods and suggests that integrating LSTM with other models could further enhance prediction accuracy.

Additionally, Vidal et al. [23] explored a hybrid CNN-LSTM model for predicting gold price volatility. This model combined the spatial feature extraction capability of convolutional layers with the temporal dependency capturing ability of LSTM layers. The results indicated a significant improvement over traditional GARCH and stand-alone LSTM models, showcasing the potential of deep learning architectures in financial market forecasting.

Furthermore, Lim et al. [24] introduced the Temporal Fusion Transformer (TFT), a state-of-the-art deep learning model specifically designed for predictive analysis of time series data. By incorporating attention mechanisms from NLP, it effectively merges attention-based architectures with recurrent neural networks and gating, yielding remarkable performance in various domains. Moreover, the TFT surpassed existing models in forecasting intraday realized volatility, while subsequent studies comparing TFT-based methods with GARCH underscore how deep learning can capture complex market dynamics [25, 26].

These studies collectively illustrate the efficacy of deep learning approaches in predicting financial market volatility. By leveraging the advanced capabilities of neural networks, these models offer improved accuracy and the ability to capture intricate patterns in financial data, providing a compelling alternative to traditional econometric methods.

Despite their advantages, deep learning approaches also have notable limitations. These models often necessitate large datasets for training, which may not always be available, especially for specific financial markets or assets. Additionally, deep learning models can be computationally intensive, requiring substantial processing power and time for both training and prediction. Another critical issue is the “black box” nature of neural networks, which complicates the interpretability of their predictions. This lack of transparency can be a significant drawback in financial contexts, where understanding the rationale behind predictions is crucial for decision-making. Moreover, these models can sometimes struggle with overfitting, particularly when dealing with highly

volatile and noisy financial data [27]. To address these challenges, ongoing research is focused on combining deep learning models with traditional methods or developing hybrid models that incorporate domain knowledge.

2.2.1 Temporal fusion transformer

Proposed by Lim et al. [24], the TFT processes time series inputs by first embedding three distinct groups of covariates into a shared latent space *without any prior knowledge on how they interact*. These three groups include observed historical features from both the target and exogenous time series, static metadata (e.g., asset sector), and known future signals (e.g., calendar events like holidays).

For each group, a small gating network learns, at every time step, which features are most relevant; those importance weights are then used to re-weight the embeddings, effectively performing dynamic feature selection as part of end-to-end training.

The re-weighted historical inputs flow into an LSTM encoder, whose hidden states capture short-term temporal dynamics; simultaneously, the static covariate embeddings pass through a context encoder and are injected into both encoder and decoder layers, allowing time-invariant characteristics (for example, an ETF's sector) to modulate the model's behavior.

Once the encoder has distilled recent patterns into recurrent representations, the decoder generates all forecast horizons in a single pass: It attends over the LSTM outputs with a customized multi-head self-attention mechanism whose shared projections ensure that the resulting attention map is both concise and directly interpretable. This attention map reveals how much each past time step contributes to each future prediction, exposing long-range dependencies without iterative back-feeding. Throughout the network, gated residual connections regulate information flow, i.e., deciding whether to pass or suppressing signals, and skip connections carry both raw embeddings and high-level features forward, safeguarding against vanishing gradients and preserving transparency of the transformation.

Finally, TFT's decoder projects its fused context and attention outputs into multiple quantiles via separate output heads trained with a pinball loss, yielding not only point forecasts but also full predictive intervals.

By combining dynamic feature gating, recurrent encoding, interpretable self-attention, static-context modulation, and quantile outputs within one unified architecture, the TFT delivers high-fidelity forecasts alongside built-in explanations of "when" and "why" each input drives the forecast at every horizon.

2.3 Hybrid models

In recent years, combining deep learning techniques with econometric models developed for financial market volatility analysis has shown significant improvement in the forecasting accuracy [28, 29]. A notable example is the model proposed by Koo and Kim [30], which integrates GARCH models with LSTM networks, employing an innovative distribution manipulation strategy to enhance stock market volatility predictions. Their results show a performance gain in terms of root-mean-square error compared to existing models, highlighting the importance of hybrid approaches for more accurate forecasts. Hajizedeh et al. [31] combined neural networks and particle swarm optimization integrating GARCH forecasted volatility to enhance Euro/Dollar exchange rate volatility forecasts. This combination improved the accuracy of predictions compared to stand-alone models, demonstrating the effectiveness of hybrid approaches in financial forecasting. Similarly, Wang and Wu [32] developed a hybrid model that merges GARCH models with LSTM networks, applying it to the Chinese financial market. Their empirical analysis demonstrated significant improvements in value-at-risk predictions, reinforcing the utility of hybrid models in complex financial settings.

Another relevant study by Kristjanpoller and Minutolo [33] specifically uses a hybrid GARCH-MLP model to forecast the volatility of gold prices. Their results indicate that incorporating an ANN with a GARCH model significantly reduces prediction errors, enhancing the precision of volatility forecasting. This study underscores

Table 1 Selected ETFs

ETF Name	Ticker	Sector
SPDR Dow Jones REIT ETF	RWR	Real Estate
Materials Select Sector SPDR	XLI	Material
Industrials Select Sector SPDR	XLI	Industry
Consumer Discretionary Select Sector SPDR	XLY	Discretionary Consumer Goods
Consumer Staples Select Sector SPDR	XLP	Basic Consumer Goods
Energy Select Sector SPDR	XLE	Energy
Financial Select Sector SPDR	XLF	Finance
Utilities Select Sector SPDR	XLU	Public Utilities
Health Care Utilities Select Sector SPDR	XLV	Healthcare
Technology Select Sector SPDR	XLK	Technology

the effectiveness of hybrid GARCH-MLP models in capturing complex patterns in financial time series that traditional models might miss.

These examples highlight the increasing importance of hybrid approaches that leverage both traditional econometric models and advanced neural networks to manage the complexities of financial data and enhance prediction accuracy. However, to the best of our knowledge, no prior work has bridged the statistical rigor of GARCH with the Temporal Fusion Transformer’s built-in feature gating, interpretable self-attention, and quantile outputs to jointly harness econometric indicators and all three covariate groups in one unified model.

3 Materials and methods

3.1 Dataset

The analysis in this study is based on historical data related to a group of ten ETFs. These instruments are investment funds that can be traded in the secondary market, offering diversification and liquidity for investors. Each of these ETFs covers a specific sector within the S&P 500 index, the major stock index in the American market. Table 1 lists the names, tickers, and reference sectors of the selected products.

We obtain the data using the open-source tool of *Yahoo Finance*,¹ covering the period from January 1, 2002, to October 31, 2023. For each asset, we collect the opening, closing, highest, and lowest daily prices, trading volume, and adjusted closing price, which accounts for dividends and stock splits. We compute the log-return from this data using Equation 14, and basic descriptive statistics are provided in Table 2. Since stock markets are closed on weekends and public holidays, the days immediately before and after a holiday were treated as consecutive in the time series. Therefore, while there may appear to be “gaps” in the calendar dates, there are no actual missing data points for any trading day: The features (open, close, high, low, volume, adjusted close) are always available for days when the market is open. Furthermore, in financial datasets, large price movements or unusual spikes in volume may appear as statistical outliers. To assess their impact, we conducted an in-depth analysis and applied the interquartile range (IQR) method, a well-known statistical technique to detect and evaluate potential outliers. The results indicate that the dataset contains only a few outliers, with virtually none exceeding the thresholds of 3 times the first quartile (Q1) or surpassing 3 times the third quartile (Q3). Given the limited presence of extreme outliers and the fact that we specifically selected non-leveraged ETFs, which are inherently less prone to extreme fluctuations, we chose to retain these observations. This decision ensures that the dataset remains as comprehensive as possible, avoiding unnecessary reductions in sample size while maintaining the integrity of the analysis.

¹ <https://help.yahoo.com/kb/SLN2311.html>.

Table 2 Descriptive Statistics I

Ticker	Mean	SD	Min	25%	50%	75%	Max
RWR	0.0112	2.0330	− 23.3883	− 0.7008	0.0693	0.7908	16.7181
XLB	0.0269	1.5535	− 13.2527	− 0.6832	0.0894	0.8277	13.1534
XLI	0.0316	1.3998	− 12.0408	− 0.5642	0.0902	0.7036	11.9127
XLY	0.0373	1.4496	− 13.5460	− 0.5594	0.1096	0.7375	9.3265
XLP	0.0324	0.9350	− 9.8668	− 0.4007	0.0677	0.5189	8.1678
XLE	0.0212	1.9838	− 22.4910	− 0.8339	0.0610	0.9709	15.2503
XLF	0.0091	2.0026	− 18.2321	− 0.7125	0.0481	0.8092	15.2394
XLU	0.0250	1.2303	− 12.0561	− 0.5410	0.0893	0.6481	12.0388
XLV	0.0376	1.1068	− 10.3817	− 0.4618	0.0741	0.6127	11.3817
XLK	0.0512	1.4498	− 14.8662	− 0.5454	0.1029	0.7334	13.0136

The ticker count is equal to 4234 for each ticker

Table 3 Descriptive Statistics II

Ticker	Kurtosis	Skewness	JB Stat.	ADF Stat.	ADF P-val	ARCH-LM Stat.	ARCH-LM P-val
RWR	18.1296	− 0.5555	58202.83	− 12.5407	2.32e−23	1215.63	6.12e−255
XLB	7.8371	− 0.3687	10931.42	− 14.8282	1.92e−27	1091.17	4.22e−228
XLI	8.6171	− 0.3972	13211.18	− 15.2983	4.35e−28	1296.18	2.55e−272
XLY	7.7453	− 0.5638	10807.36	− 16.1112	5.01e−29	1040.54	3.45e−217
XLP	11.2881	− 0.4317	22610.61	− 21.4045	0.0	1473.73	1.19e−310
XLE	13.0419	− 0.6947	30347.39	− 69.5694	0.0	1019.47	1.19e−212
XLF	14.3405	− 0.1748	36301.51	− 11.2507	1.71e−20	1076.35	6.63e−225
XLU	14.1525	0.0137	35335.05	− 16.2537	3.62e−29	1532.46	0.0
XLV	10.7545	− 0.3366	20484.30	− 17.2865	5.74e−30	1447.98	4.33e−305
XLK	9.5439	− 0.2268	16105.49	− 16.0202	6.22e−29	1077.29	4.15e−225

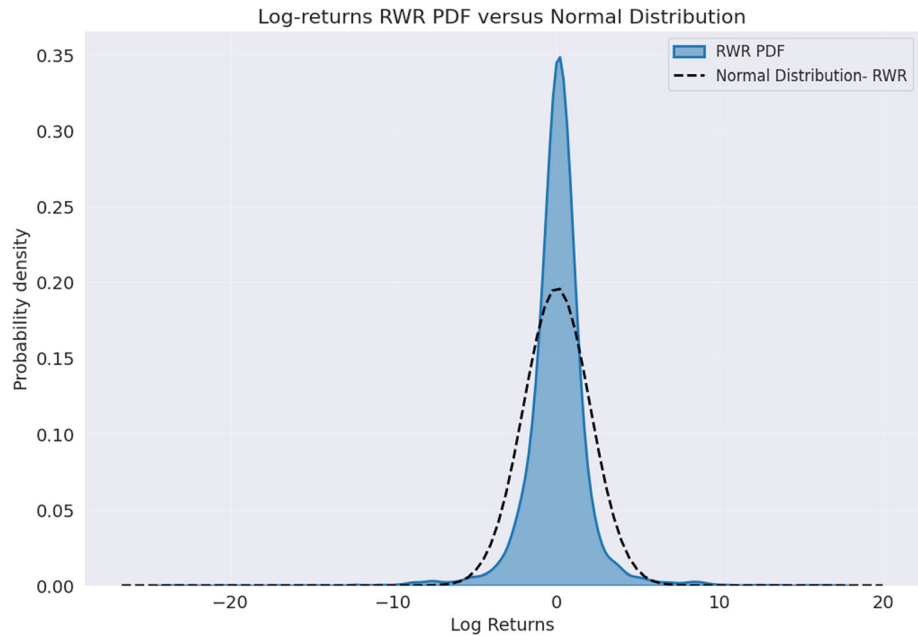
For each ticker, the p value of the JB statistics resulted <0.001

Strategic insights can be gleaned from Table 3, which illustrates that the log-returns of all the tickers exhibit a pronounced kurtosis, with values significantly exceeding 3, indicating leptokurtic distributions. This suggests a substantial presence of extreme values, both positive and negative, compared to a normal distribution. For example, RWR displays an exceptionally high kurtosis value (18.13), indicating a concentration of values around the mean and a heavy tail, as depicted in Fig. 1.

The skewness values are predominantly close to zero and negative, indicating a slight leftward asymmetry in the distributions, except for the XLU ticker. This implies that there are more extremely negative returns than positive ones. High ARCH-LM and JB statistic values, along with low p values, respectively, indicate the presence of conditional heteroskedasticity and the rejection of the normality assumption for the return distribution of all tickers. Similarly, high augmented Dickey–Fuller statistic values and low p values suggest that the return time series are not unit root processes, indicating that they are stationary. In conclusion, these data reflect common characteristics of stock returns, as previously introduced in Sect. 3.2, such as non-normality, asymmetry, heavy tails, leptokurtosis, and time-varying volatility. Furthermore, the results of ADF and ARCH-LM tests facilitate the use of generalized autoregressive conditional heteroskedasticity (GARCH) models. In fact, the choice of this family of models is justified by the presence of previously verified heteroskedasticity. Moreover, before proceeding with the training of the deep learning models the dataset was enriched by adding auxiliary data, such as:

- **Market data** daily closing price values of the VIX, an index measuring investors' expectations of future volatility, as well as the prices of gold, oil, the S&P500 itself; the US Treasury bonds with 2-year and 10-year maturities and the yield curve; US Dollar/ EUR and US Dollar/Yen exchange rates;

Fig. 1 Logarithmic returns
RWR PDF vs Normal
Distribution



- **temporal features**, known at the time of forecasting, such as month day, weekday number, week number in month, year and month itself;
- **static features** that can be managed by the considered architecture, such as the product ticker and its sector of belonging.

3.2 Volatility proxies

Financial markets play a fundamental role in a country's economic development. Since participants in these markets are concerned, or should be, about the risks associated with the activities they invest in, modeling and forecasting the volatility of financial assets is of primary importance. For a given financial asset, its behavior can be modeled using the following stochastic differential equation (SDE):

$$d\log P_{i,t} = \mu_i dt + \sigma_{i,t} dW_t, \quad (11)$$

in which P_i represents the price of the i -th financial asset, $d\log P_{i,t}$ its differential variation over a small temporal interval dt , expressed as the combination of both a deterministic term $\mu_i dt$ and a stochastic one $\sigma_{i,t} dW_t$. Within this context, μ_i is known as *drift* and represents the expected rate of growth or decrease of the price per unit of time, while the stochastic component $\sigma_{i,t}$ accounts for the volatility or random fluctuations in the asset's price: dW_t is a differential variation of a Wiener process (Brownian motion), representing the random fluctuations in price.

Following the past equation, volatility can therefore be defined as an instantaneous variable that scales a stochastic process, quantifiable through a time integral to obtain the so-called integrated volatility of asset i during the interval $(t-s, +t]$:

$$IV_{i,t}(h) = \int_{t-h}^t \sigma_{i,s}^2 ds \quad (12)$$

where h expresses the length of the time window being considered (either expressed in seconds, minutes, hours, days, or months, depending on the frequency of interest). However, since financial assets are discrete by nature,

the previous equation is difficult to quantify directly and must be thus approximated. In this sense, different proxies, i.e., alternative measures or estimators that serve as substitutes for the true but unobservable integrated volatility can be considered, depending on the purpose and context in which volatility is analyzed. For instance, one of the most commonly used proxies for approximating Eq. 12 is the realized volatility (RV), defined as the square root of the sum of squared returns within the time window $[\tau_1, \tau_2]$:

$$RV = \sqrt{\sum_{t=\tau_1}^{\tau_2} r_t^2} \quad (13)$$

with

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right) \quad (14)$$

where r_t represents the return of the asset at time t , defined as the logarithmic difference between the prices at time t and $t - 1$. However, this proxy is considered when data with sufficiently high sampling frequency (5/15 min) are available, which does not fit the ETFs we analyze in this work. Therefore, here we employ a second proxy, the HV, visually represented for each ETF in A, which quantifies the dispersion in daily closing prices of an asset. It is defined as the standard deviation of the series of logarithmic returns calculated over a certain time window:

$$HV = \sqrt{\frac{1}{N} \sum_{t=\tau_1}^{\tau_2} (r_t - \mu_{\tau_1, \tau_2})^2} \quad (15)$$

where r_t is still the logarithmic returns and μ_{τ_1, τ_2} represents their average in the considered time window. From a statistical point of view, the previous equation represents an estimate based on N observations and is *biased downward* because it considers deviations from the sampled arithmetic mean and not from the true expected value $E(r)$, which is unknown, and thus introduces an estimation error. This introduced bias can be eliminated by multiplying the previous equation by the factor $n/(n - 1)$, obtaining the following corrected estimate:

$$\hat{HV} = \sqrt{\frac{1}{N-1} \sum_{t=\tau_1}^{\tau_2} (r_t - \mu_{\tau_1, \tau_2})^2} \quad (16)$$

Within this work, a third proxy, which well adapts to the data at hand, is considered, the Garman–Klass Volatility [34]:

$$GK = \sqrt{\frac{1}{N} \sum_{i=1}^N \left[0.5 \times \left(\ln \frac{H_i}{L_i} \right)^2 - (2 \ln 2 - 1) \times \left(\ln \frac{C_i}{O_i} \right)^2 \right]} \quad (17)$$

where H_i is the daily high price, L_i is the daily low price, C_i is the daily closing price, and O_i is the daily opening price. Introduced by Mark Garman and Michael Klass, it is particularly known for its ability to provide a more accurate measure of volatility than simpler models. This approach captures more information about price movements, particularly in markets with significant intraday fluctuations, leading to a more nuanced understanding of an asset's volatility.

Nevertheless, while HV and GK serve as effective realized volatility proxies, their applicability varies across different market conditions. HV, being a backward-looking measure, may fail to immediately capture abrupt

volatility shifts, particularly during financial crises. GK, while improving estimation by incorporating intraday price fluctuations, assumes a continuous market and may underestimate volatility in discontinuous trading environments. However, the use of Exchange Traded Funds (ETFs) in our study provides a natural hedge against excessive short-term fluctuations, as ETFs aggregate multiple assets, reducing the impact of extreme market movements on individual proxies.

Therefore, in this work, we utilized both HV and GK to evaluate the accuracy of the model predictions, allowing us to cross-validate the results and ensure robustness in our volatility assessments.

3.3 Proposed model

Our proposed model, GARCH-TFT, represents an innovative hybrid architecture that seamlessly integrates the robust volatility modeling capabilities of GARCH models with the advanced features of the TFT [24]. This hybrid design leverages the precision of GARCH models in handling volatility alongside the dynamic forecasting capabilities of TFT to address a broad spectrum of time series prediction challenges. This combination allows to overcome the known limits of GARCH models with a structure, the TFT, that is more flexible across data inputs, temporal dynamics, and scalability. Table 4 reports the main aspects of comparison between the two kinds of models.

The GARCH component of the proposed model employs a family of models, including GARCH, E-GARCH, T-GARCH, and GJR-GARCH, to process financial log-returns. These models are calibrated using historical data, spanning a five-year window, with a skewed-t distribution selected according to the Akaike information criterion (AIC), a standard criterion that measures the relative quality of statistical models for a given dataset [35]. The GARCH layer outputs key features such as residuals, time t -conditioned volatility, and predicted volatilities for the next 10 days. These features enrich the set of input features used to train the TFT component.

The TFT component incorporates the GARCH-derived features, i.e., the combined outputs (residuals, time-conditioned volatility, and predicted volatility) of the four GARCH models employed in the first block of the workflow, alongside a rich set of auxiliary data as depicted in Fig. 2, including:

- Market data, such as daily closing prices of the VIX, gold, oil, and the S&P500 index; US Treasury bond yields for 2-year and 10-year maturities; and currency exchange rates (USD/EUR, USD/JPY).
- Temporal attributes, including the day of the month, day of the week, week number within the month or year, and the month itself.
- Static features, such as the product ticker and its sector of operation.

The TFT dynamically learns temporal dependencies and feature importance via its attention mechanisms. It is trained to predict two proxies for volatility: HV, based on price changes over a 21-day window, and GK, which incorporates high, low, and closing prices.

Table 4 Comparison of GARCH and TFT models

Aspect	GARCH	TFT
Assumptions	Relies on strong parametric assumptions and linear relationships	Data-driven approach without requiring a predefined functional form
Data Input	Primarily uses logarithmic returns, which may not fully capture the complex span of volatility-driving factors	Integrates diverse auxiliary inputs (e.g., VIX levels, commodity prices, bond yields, temporal attributes), , allowing for a more comprehensive volatility modeling approach
Temporal Dynamics	Assumes a constant structure over time, making it less adaptive to sudden changes	Dynamically learns temporal dependencies and adjusts weighting to capture regime changes
Scalability	Requires asset-specific parameter estimation, with the need for recalibrating individual models	Can learn across multiple assets within a single training framework, enhancing computational efficiency

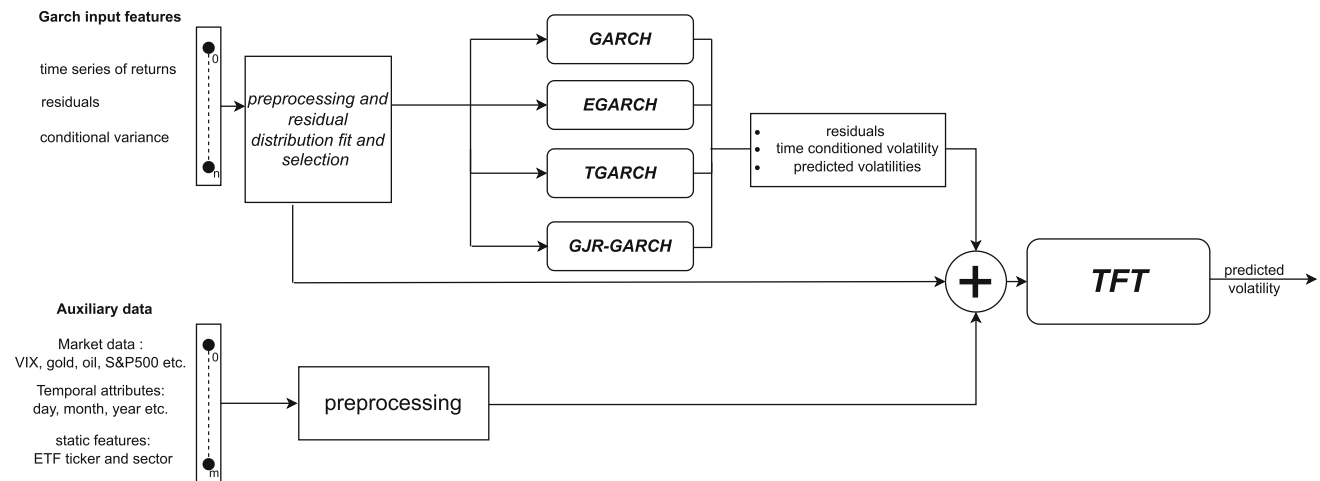


Fig. 2 Schematic workflow of the Proposed GARCH-TFT Hybrid Model. The diagram illustrates how GARCH-derived features are integrated with various auxiliary data (including market data, temporal attributes, and static features) to form the input for the Temporal Fusion Transformer

The hybrid nature of the GARCH-TFT model lies in the synergy between its components. The GARCH layer provides a robust statistical foundation by modeling the underlying market dynamics, while the TFT layer enriches this foundation with contextual information and advanced pattern recognition capabilities.

3.4 Experimental setup

The experimental setup for this study is organized into two distinct phases.² In the first phase, we calibrated a family of GARCH models using log-returns. This involves estimating the models' parameters with historical market data spanning five years, from January 1, 2002, to January 1, 2007. For the residuals, we employed a skew-t distribution, which resulted as the best fit, after a comprehensive analysis that evaluated different time periods (ranging from three to eight years), among various distributions including skew-t, t, normal, and GED. This analysis was conducted with the Akaike information criterion (AIC), which measures the relative quality of statistical models for a given dataset. Among all the assessed distributions, the skew-t one provided the best balance between model fit and complexity, making it the optimal choice for our residuals, ensuring a more accurate estimation of volatility dynamics, ultimately improving the robustness of our forecasting models.

Once the GARCH models were calibrated, we proceeded to the second phase: training the TFT. This second phase utilized an online training paradigm, where the model is continuously updated with new data as it becomes available. Specifically, from January 1, 2007, to October 30, 2023, we conducted daily updates for each market day and each econometric model. During this period, the following values were extracted from the calibrated GARCH models and utilized for training: the residuals and time t -conditioned volatility, as well as the model's predicted volatility values for the next ten days. The selection of this extensive time frame for prediction serves to ensure robustness and reliability in our forecasts. Additionally, it aligns with the requirements outlined in the Basel II agreement, which emphasizes the importance of accurate risk assessment and management over extended periods.

These data, together with the previously described datasets, constitute the training set for the GARCH-TFT model. Importantly, all numerical features were normalized using statistics derived exclusively from the training set to prevent any information leakage and minimize overfitting. Moreover, the validation set employed for the

² The reader can find more implementation details, data, and codes that can be reused on our repository associated with this work: <https://github.com/lbacco/garch-tft>.

Table 5 Dataset splitting periods

Split	Period
Train	January 1, 2007–December 31, 2021
Val	January 1, 2022–December 31, 2022
Test	January 1, 2023–October 30, 2023

hyperparameters tuning was also utilized both for halting the training once the performance on this unseen dataset plateaued or began to degrade. This regularization technique, called early stopping, is often employed to reduce the overfitting by avoiding the model to lose generalizability on unseen datasets by further specializing on the training set. Each variant of the model was trained using two different volatility proxies: HV and GK, both calculated over a 21-day market window. Accordingly, we refer to these models as GARCH-TFT_{HV} and GARCH-TFT_{GK} throughout this paper. We then compared these hybrid models with their non-hybrid counterparts based on GARCH and TFT, as well as with two other hybrid models from the literature: GARCH-MLP (see Table 11) and GARCH-LSTM (see Table 12). For consistency, these comparison models are also denoted with a subscript indicating the volatility proxy used during their training.

In this scenario, it is worth considering that while the latter models require individual training for each ETF, both the TFT and GARCH-TFT are global models that, with a single training session, provide an estimation for multiple assets under examination. This characteristic also has an impact on the type of dataset required for the network. Specifically, in the case of the transformer model, the dataset comprises 43,000 observations. For the same index and date, there are 10 different observations corresponding to different tickers. This differs from the deep models, where each stock asset has its own model trained on the corresponding dataset. In both cases, the available dataset is divided as shown in Table 5.

We trained the model on a dataset that spans a range of significant economic events, covering the period starting from the subprime mortgage crisis, followed by the subsequent recovery, and leading up to the global economic instability caused by the COVID-19 pandemic in 2020. The model was then validated during a year when the war in Ukraine had a major impact, especially on global markets. Finally, it was tested over three quarters of 2023, a period marked by rising inflation and the consequent increase in interest rates, which, as usual, influenced investors' decisions. We designed this validation protocol to stress-test the model across a historically unstable period.

The validation set serves not only to prevent overfitting of the models through EarlyStopping but also to identify the optimal set of hyperparameters. In order to tune the hyperparameters of the model, we systematically explored 2,646 unique hyperparameter combinations (Table 6) via a grid search methodology, training and evaluating each model on the validation set to select the best-performing configuration (Table 7).

The best resulting hyperparameters are presented in Table 7 for both the target variable and TFT variants. For a comprehensive list of possible choices, please refer to Table 6.³ It is important to consider that, despite the extensive neural architecture search process requiring significant computational resources and time, training the final network with the optimal hyperparameters took approximately two hours on a machine equipped with two Tesla K40 GPUs. This makes it feasible even on a non-dedicated computer. Moreover, a single inference of the complete GARCH-TFT model can be executed in under a millisecond.

The input window length is set to 15 market days, equivalent to 3 weeks. This choice is strategically made to capture a short-term market perspective while striking a balance between utilizing recent data and avoiding excessive noise. This duration allows the model to incorporate the latest market trends and subtle shifts, ensuring a blend of timely responsiveness and data stability.

Moreover in order to evaluate the importance of individual features in our model, the attention mechanism within the TFT was utilized. This approach allowed for a detailed examination of each feature's contribution

³ To access the list of hyperparameters for the MLP and LSTM models, please consult Appendix B and Appendix C, respectively.

Table 6 Hyperparameters choices

Parameters	Options
Dropout Rate	[0.1, 0.2, 0.3, 0.4, 0.5, 0.7, 0.9]
Hidden Layer Size	[10, 20, 40, 80, 160, 240, 320]
Minibatch Size	[64, 128, 256]
Learning Rate	[1e-4, 1e-3, 1e-2]
Max Gradient Norm	[0.01, 1.0, 100.0]
Heads Number	[1, 2, 3, 4]

Table 7 Hyperparameters optimization results

Best Hyperparameter				
Parameters	TFT _{HV}	TFT _{GK}	GARCH-TFT _{HV}	GARCH-TFT _{GK}
Dropout Rate	0.2	0.2	0.4	0.1
Hidden Layer Size	320	40	240	320
Minibatch Size	64	256	64	256
Learning Rate	0.01	0.01	0.01	0.01
Max Grad. Norm	0.01	1	1	1
Heads Number	1	2	2	4

during the training process. By analyzing the attention weights, we were able to determine which features significantly influenced the model's performance, thereby shedding light on the predictive dynamics.

The accuracy of the forecasting models is evaluated using two standard error metrics: the root-mean-square error (RMSE) and the mean absolute error (MAE). Mathematically, RMSE is the square root of the average of the squared errors. Mathematically, it can be represented as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (18)$$

RMSE measures the average deviation of the predicted values from the actual values. MAE, on the other hand, represents the average of the absolute differences between predicted values and actual values:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (19)$$

where y_i are the actual values, $\hat{y}_i$ are the predicted values, and n is the number of observations. MAE provides an indication of the average magnitude of errors in the forecasts.

While the former penalizes larger errors more heavily due to the squaring of each error term, which can be particularly useful in contexts where large deviations are particularly undesirable, the latter provides a straightforward average of absolute errors, making it easier to interpret and less sensitive to outliers. A lower RMSE value indicates more accurate predictions, while a higher value suggests large discrepancies between predictions and reality. A low MAE value suggests that the predictions are close to the actual values, while a high value indicates large discrepancies. One of the main advantages of MAE is its ease of interpretation, as it directly measures the average error in units of the variable of interest. By employing both MAE and RMSE, we aim to provide a more comprehensive overview of our results and facilitates comparison with future works.

Finally, thanks to the attention layer embedded within the TFT architecture, it was possible to conduct a detailed analysis of the significance of the features inputted into the model. This analysis provided valuable insights into the hierarchy of the input features and their respective weights within the model. The attention

mechanism facilitated a nuanced understanding of how different features influence the model's predictions, highlighting the importance of specific variables in forecasting volatility.

To assess the robustness of our volatility forecasts, we implement an error propagation analysis by introducing controlled noise into the input data and evaluating its impact on model predictions. This approach quantifies the sensitivity of the GARCH-family models to perturbations in the log-return time series before passing the generated forecasts to the TFT for the final error evaluation.

The log-return series is perturbed with normally distributed noise, where the mean is set to zero, and the standard deviation is equal to the empirical standard deviation of the log-returns. For each rolling window, we fit the following volatility models on both the original and noisy log-return series: GARCH, E-GARCH, T-GARCH, GJR-GARCH. To evaluate the impact of noise, we compute the absolute mean percentage deviation between forecasts generated from the original and the noisy data:

These errors are stored alongside the original residuals and conditional volatilities, providing insight into each model's sensitivity to perturbations.

After deriving volatility forecasts and quantifying model sensitivity, the resulting forecast vectors are passed to the TFT model for further percentual error evaluation. This step ensures that error propagation is analyzed not only at the level of traditional econometric models but also within the deep learning part of the proposed framework, thereby reinforcing the robustness of the overall predictive system.

4 Results and discussion

The following tables showcase the models' results. Table 8 presents the comparison between the performances of considered model in terms of RMSE, while Table 9 reports the results in terms of MAE. Statistical independence tests conducted using Wilcoxon's method confirm that the outputs of all examined models are statistically independent, reinforcing the validity of the comparisons.

The TFT and GARCH-TFT models consistently demonstrate superior performance, as indicated by their lower RMSE and MAE scores across most tickers. This is especially evident in the HV columns, where these models often have the lowest error rates. Traditional GARCH models, including T-GARCH, E-GARCH, and GRJ-GARCH, generally show higher RMSE and MAE values, suggesting that they are less accurate in forecasting volatility compared to the hybrid and deep learning models, in respect to considered proxies. However, it is important to highlight how the GARCH model has the lowest errors between other econometrics models. The TFT model appears particularly effective in HV forecasting. Its performance in GK (Garman–Klass) forecasting is also commendable, though GARCH-TFT occasionally outperforms it. The GARCH-TFT hybrid model excels in both GK and HV forecasting for various tickers, highlighting the benefits of combining traditional econometric models with advanced deep learning techniques. The GARCH-MLP and GARCH-LSTM models show mixed results, performing better if compared to the stand-alone GARCH models but not as consistently as TFT or GARCH-TFT. Different models exhibit varying degrees of effectiveness depending on the ticker. For example, XLK and XLE show higher RMSE and MAE values across all models, indicating these tickers' volatilities are more challenging to predict accurately. In contrast, models tend to perform better with tickers like XLP and XLV, as indicated by the generally lower RMSE and MAE values. The results suggest that the TFT and GARCH-TFT models could offer more reliable volatility forecasts, which is crucial for risk assessment and portfolio optimization. To provide visualization example, we have randomly chosen the XLU ticker (neither the best nor the worst performer) to illustrate the forecasted volatility versus the actual observed volatility over a 10-day period for the XLU ticker, comparing different models (Figs. 3 and 4).

As can be seen in Fig. 3, the TFT and GARCH-TFT models track the historical line more closely than GARCH, indicating a better performance in predicting the selected proxy. This visualization supports the quantitative findings and provides a clear comparison between the different modeling approaches. Figure 4 further underscores the comparative accuracy of deep models and TFT variants. The clear delineation of

Table 8 RMSE results for both GK and HV proxies

Ticker	GARCH		T-GARCH		E-GARCH		GRJ-GARCH		TFT		GARCH-TFT		GARCH-MLP		GARCH-LSTM	
	GK	HV	GK	HV	GK	HV	GK	HV	GK	HV	GK	HV	GK	HV	GK	HV
RWR	0.311	0.192	0.381	0.228	0.350	0.226	0.357	0.204	0.135	0.167	<i>0.086</i>	0.176	0.180	0.442	0.313	0.200
XLB	0.369	0.239	0.432	0.292	0.381	0.243	0.413	0.274	0.178	0.151	<i>0.089</i>	0.146	0.135	0.355	0.249	0.215
XLI	0.352	0.247	0.409	0.297	0.387	0.274	0.353	0.242	0.186	0.126	<i>0.081</i>	0.111	0.239	0.166	0.453	0.333
XLY	0.395	0.189	0.367	0.187	0.422	0.212	0.403	0.190	0.190	0.118	<i>0.080</i>	0.139	0.198	0.135	0.248	0.193
XLP	0.201	0.185	0.280	0.264	0.230	0.171	0.364	0.153	0.144	0.082	<i>0.064</i>	0.092	0.120	0.570	0.092	0.302
XLE	0.466	0.280	0.498	0.297	0.518	0.320	0.467	0.297	0.177	0.230	<i>0.202</i>	0.235	0.274	0.312	0.312	0.235
XLF	0.453	0.355	0.517	0.391	0.446	0.359	0.544	0.404	0.208	0.225	<i>0.150</i>	0.237	0.359	0.358	0.440	0.730
XLU	0.161	0.194	0.196	0.231	0.178	0.202	0.203	0.234	0.135	0.179	<i>0.128</i>	0.176	0.232	0.175	0.175	0.269
XLV	0.228	0.199	0.310	0.290	0.248	0.214	0.239	0.188	0.123	0.079	<i>0.081</i>	0.113	0.181	0.214	0.215	0.102
XLK	0.486	0.264	0.563	0.362	0.528	0.301	0.543	0.308	0.171	0.126	<i>0.078</i>	0.131	0.135	0.443	0.127	0.848

For each of the two proxies, the best performance is highlighted in *italics* and **bold**, respectively

Table 9 MAE results for both GK and HV proxies

Ticker	GARCH		T-GARCH		E-GARCH		GRJ-GARCH		TFT		GARCH-TFT		GARCH-MLP		GARCH-LSTM	
	GK	HV	GK	HV	GK	HV	GK	HV	GK	HV	GK	HV	GK	HV	GK	HV
RWR	0.279	0.162	0.344	0.190	0.325	0.194	0.305	0.172	0.096	0.118	<i>0.065</i>	0.135	0.156	0.402	0.284	0.154
XLB	0.345	0.206	0.400	0.252	0.352	0.209	0.380	0.234	0.122	0.113	<i>0.064</i>	0.111	0.113	0.322	0.222	0.183
XLI	0.331	0.213	0.390	0.267	0.366	0.244	0.332	0.210	0.132	0.097	<i>0.064</i>	0.087	0.205	0.139	0.424	0.297
XLY	0.368	0.148	0.330	0.150	0.400	0.171	0.364	0.153	0.144	0.092	<i>0.064y</i>	0.106	0.120	0.570	0.216	0.162
XLP	0.190	0.172	0.266	0.250	0.219	0.202	0.208	0.191	0.155	0.059	<i>0.060</i>	0.079	0.178	0.088	0.076	0.253
XLE	0.427	0.238	0.461	0.256	0.474	0.271	0.417	0.230	0.135	0.176	<i>0.099</i>	0.186	0.190	0.225	0.282	0.198
XLF	0.429	0.331	0.457	0.357	0.421	0.329	0.478	0.372	0.160	0.131	<i>0.102</i>	0.148	0.343	0.307	0.415	0.693
XLU	0.128	0.151	0.168	0.183	0.148	0.160	0.164	0.179	0.094	0.117	<i>0.087</i>	0.120	0.141	0.171	0.124	0.211
XLV	0.213	0.185	0.299	0.270	0.234	0.204	0.230	0.200	0.138	0.064	<i>0.048</i>	0.093	0.067	0.217	0.197	0.087
XLK	0.453	0.227	0.520	0.308	0.498	0.267	0.468	0.263	0.171	0.103	<i>0.061</i>	0.099	0.119	0.418	0.096	0.818

For each of the two proxies, the best performance is highlighted in *italics* and **bold**, respectively

forecasting lines offers a visual testament to the robustness of the TFT and GARCH-TFT models, particularly when compared to the historical volatility trend line. Therefore, the visual data align with our initial findings, highlighting the enhanced reliability of the TFT and GARCH-TFT models.

Given the success of the TFT-based models, we examine which historical and future features have been crucial for this kind of models. In particular, we have focused on the TFT_{HV} , which achieved the lowest MAE on the HV proxy among all the investigated models, and interpretability studies were performed by analyzing the set of the interpretable multi-head attention weights. The model's attention weights have been aggregated across all 1830 samples, representing the attention distribution over the 15 encoder time steps for each of the 26 features. To gain a deeper understanding of these distributions, we computed the median attention weight for each feature across all time steps and samples. This computation gives us insight into the average level of attention the model pays to each feature, both in historical and future contexts. Figures 5 and 6 visually represent these median attention weights for historical and future input features, respectively.

Figure 5 delineates the significance of each historical feature in the model's decision-making process. The model places substantial emphasis on *Volatility10*, indicating that the 10-day historical volatility is the most

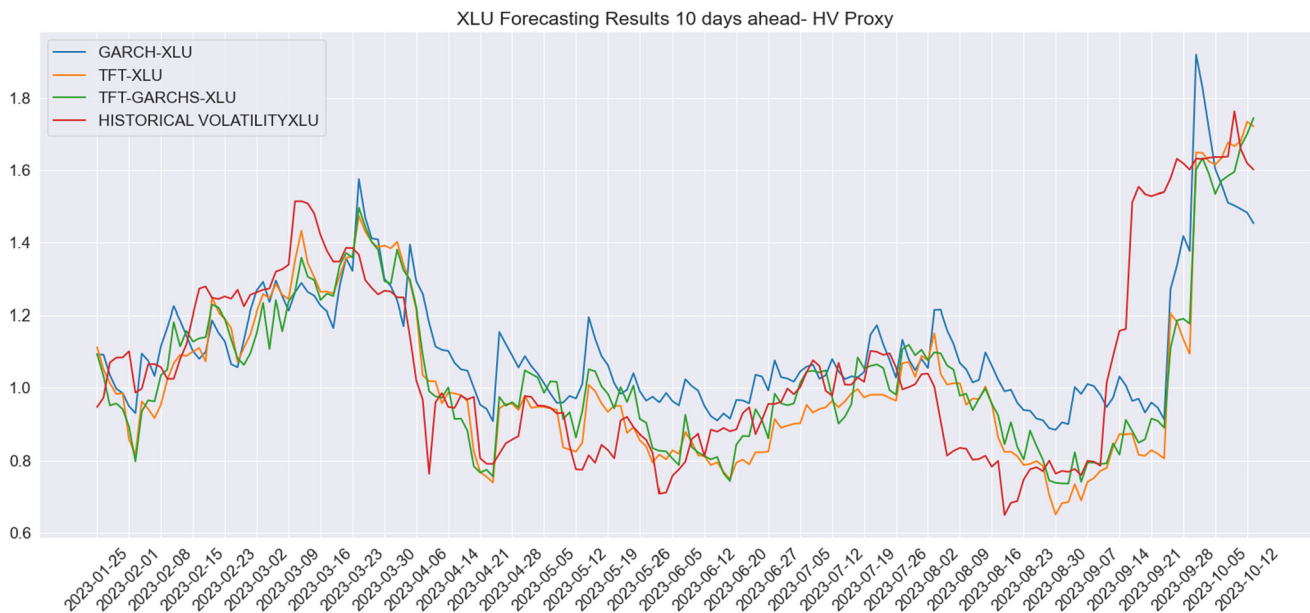


Fig. 3 XLU comparative forecasting results I

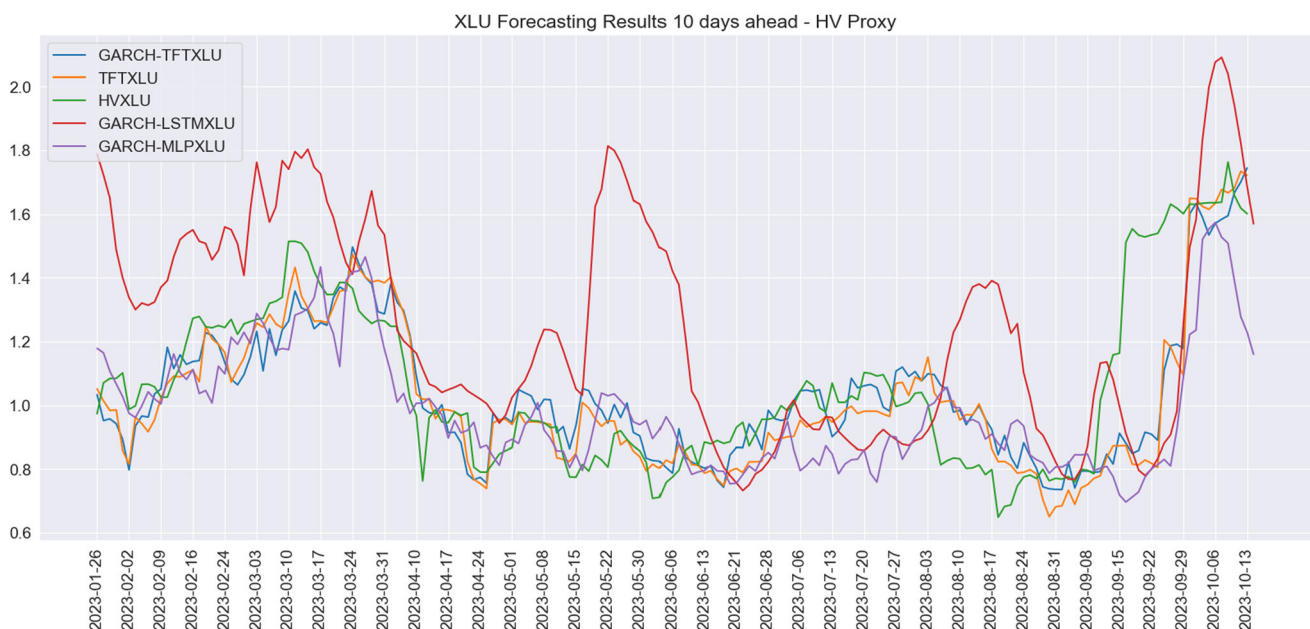


Fig. 4 XLU comparative forecasting results II

influential feature, followed closely by *HV Vol*. This highlights the model's reliance on short-term volatility measures. The *open_to_close* feature, which represents the daily price range, also has a notable impact, reflecting the importance of daily price movements in forecasting. Other features provide additional but progressively lesser contextual information to the model. This suggests that while the model considers a broad range of historical data, it prioritizes features directly related to price volatility and movement.

Conversely, Fig. 6 shows the relative importance of time-related features for future predictions. *Days_from_start* is the most critical feature, indicating that the model places significant weight on the overall time elapsed. The cyclic temporal features like *month* and *week_of_year* also have considerable importance, which implies that the model finds regular patterns in monthly and weekly cycles relevant for prediction. In contrast,

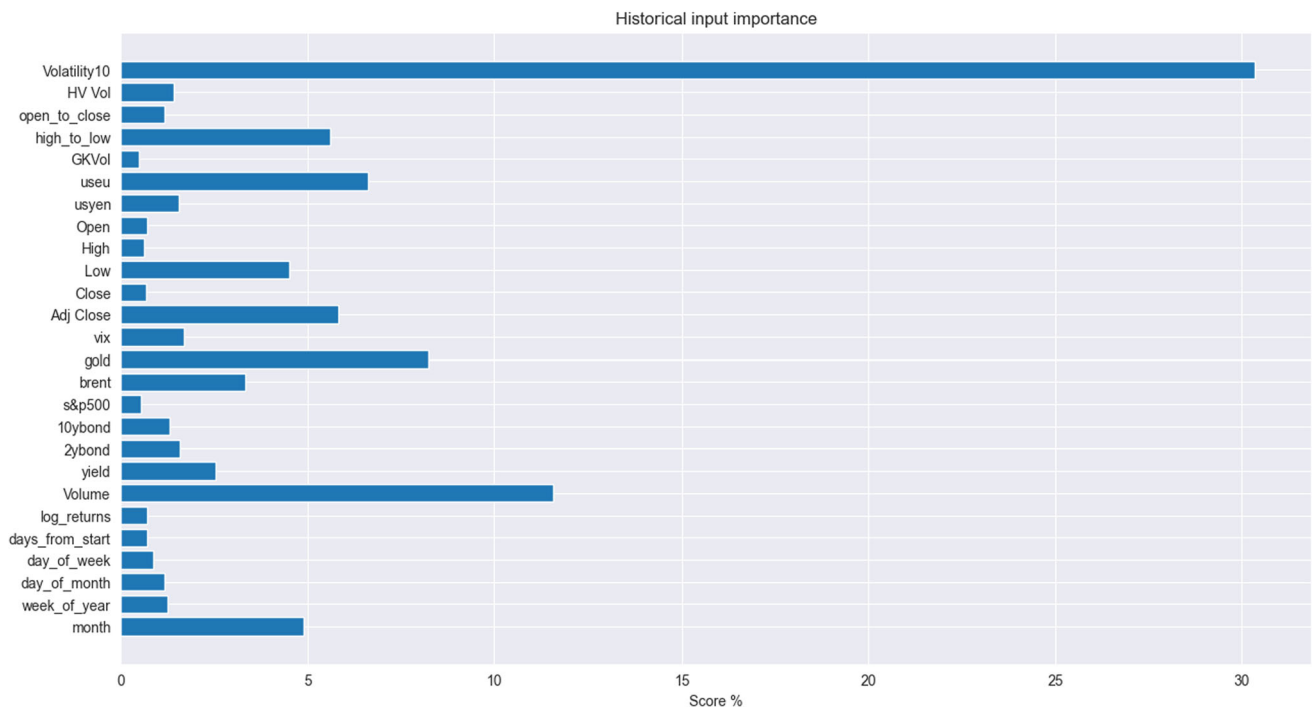


Fig. 5 Historical features input importance

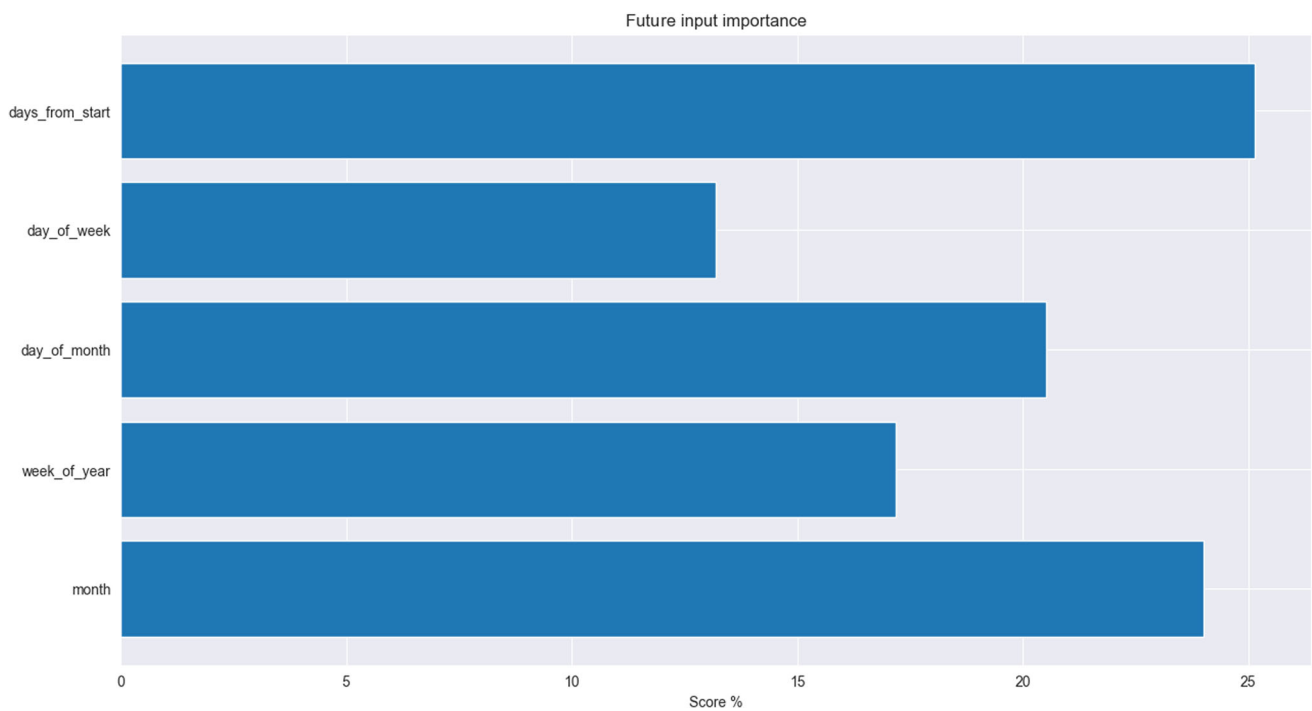


Fig. 6 Future input importance

day_of_week and *day_of_month* have less influence, suggesting that the specific day within a week or month is of lower significance for the model's forecasts.

Together, these graphs offer a compelling visualization of how the model prioritizes different aspects of the data. Historical volatility and price changes are critical for understanding past market behavior, while the elapsed

Table 10 MAE percentage and RMSE percentage for different volatility models across ETFs

Ticker	GARCH		EGARCH		TARCH		GJR		TFT	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
RWR	0.8743	0.9007	0.8590	0.8898	0.8571	0.8876	0.8672	0.8986	0.8450	0.8720
XLB	0.8276	0.8454	0.8209	0.8490	0.7827	0.8178	0.7977	0.8228	0.7705	0.8050
XLI	0.8744	0.8895	0.8474	0.8719	0.8368	0.8653	0.8728	0.8938	0.8250	0.8600
XLY	0.8079	0.8370	0.8030	0.8382	0.8410	0.8741	0.8063	0.8376	0.7900	0.8200
XLP	0.6106	0.6196	0.5842	0.5983	0.5568	0.5749	0.6038	0.6151	0.5400	0.5600
XLE	1.3584	1.3858	1.3239	1.3619	1.3383	1.3731	1.3498	1.3816	1.2900	1.3200
XLF	0.9847	1.0149	0.9938	1.0304	0.9729	1.0196	0.9695	1.0097	0.9400	0.9800
XLU	0.7409	0.7617	0.7266	0.7522	0.7077	0.7331	0.7093	0.7354	0.6900	0.7200
XLV	0.7102	0.7177	0.6866	0.7012	0.6431	0.6610	0.6950	0.7051	0.6200	0.6400
XLK	0.9567	0.9810	0.9193	0.9502	0.8955	0.9316	0.9450	0.9736	0.8800	0.9100

time and cyclical patterns are essential for future projections. Such insights can be instrumental in refining the model further and selecting the most informative features for effective financial forecasting.

The results in Table 10 address the error propagation analysis described in Sect. 3.4 highlighting the relative performance of the different models in terms of MAE percentage and RMSE percentage across various ETFs. Among the GARCH-family models, EGARCH and TARCH generally exhibit lower error values, indicating their ability to better capture volatility dynamics compared to standard GARCH(1,1). However, as shown in the final column, the TFT model achieves slightly lower error rates overall, demonstrating its capacity to refine volatility predictions despite being influenced by the noise introduced by the GARCH-based forecasts. This outcome aligns with the interpretability analysis of the TFT, which revealed that its predictions are primarily driven by the volatility estimates provided by the GARCH-family models. Nevertheless, the observed reduction in MAE and RMSE can be attributed to the additional information extracted from the other input features, reinforcing the role of TFT in leveraging broader market dynamics to enhance volatility forecasting accuracy.

5 Conclusion

In this work, we explore the predictive capabilities of a TFT and of an hybrid GARCH-TFT model for forecasting the volatility of returns on ten ETFs, with HV and the GK method that serve as our volatility proxies. Our findings reveal that the hybrid GARCH-TFT model significantly surpasses other examined models in estimating the GK proxy and competes closely with the stand-alone TFT for HV estimation, demonstrating the effectiveness of combining econometrics GARCH models with advanced deep learning techniques, such as the TFT.

Such advancements in forecasting methodologies highlight the potential for enhancing predictive accuracy in financial markets, a crucial aspect for assisting managers and investors for risk management and investment decision-making. By providing more accurate predictions of market volatility, this model helps to better navigate market fluctuations, enhancing their strategic planning and investment performance, and thus in optimizing portfolio allocation, reducing risks, and maximizing returns. Policymakers can also benefit from the insights provided by the integration of the standard econometrics models with the TFT. Accurate volatility predictions are essential for evaluating the potential impacts of regulatory interventions and for designing policies aimed at stabilizing financial markets. By understanding the dynamics of market volatility more precisely, policymakers can make informed decisions to mitigate systemic risks, enhance market resilience, and promote economic stability.

While the results presented in Tables 8 and 9 may not appear substantially strong from a purely performance-based perspective, it is essential to evaluate whether these improvements translate into economically meaningful gains for investors. The results in tables should be interpreted within the context of the unique historical period

analyzed, specifically the years surrounding the COVID-19 pandemic, which was characterized by unprecedented financial market volatility. This period was heavily influenced by macroeconomic disruptions and governmental interventions, which significantly impacted stock returns.

Despite these challenges, the model demonstrates a strong ability to focus on key economic indicators, including the Euro-Dollar exchange rate, gold prices, and crude oil prices (Brent, as shown in Fig. 5). Furthermore, a substantial portion of the attention mechanism is directed toward interest rates (2-year and 10-year bond yields, as illustrated in Fig. 5), highlighting the model's responsiveness to critical macroeconomic variables. The ability of our approach to incorporate these factors during periods of market stress underscores the robustness and adaptability of the proposed framework, reinforcing its relevance for financial decision-making.

Looking ahead, despite the promising results, a few limitations could be handled in future works. For instance, the experimental phase is centered on ETFs within the US market. This focus may limit the generalizability of the findings to other geographic regions and markets, and to other assets such as individual stocks, bonds, or cryptocurrencies, which may exhibit different volatility dynamics. For instance, other econometrics models can be applied to assets like commodities, since they exhibit price dynamics influenced by factors not present in stock markets, e.g., supply and demand shocks, seasonal patterns, and geopolitical events, which GARCH models might not effectively capture.

Among the potential future directions for this work, there is also the development of a web application that integrates these volatility estimation methods. This platform could serve as a valuable tool for retail investors, aiding in their investment decisions by incorporating features like transaction cost considerations, which are particularly relevant for this investor class. Nevertheless, a deeper exploration of feature significance and interpretability in diverse market conditions would enhance our contribution. In future work, we aim to extend our analysis to additional time periods and refine our model, which we believe could lead to even greater performance improvements. Moreover, given the dynamic nature of time series research, evaluating the performance of other Transformer-based models on volatility forecasting presents an exciting avenue for further exploration.

Appendix A: Historic volatility

In order to improve the clarity of the paper, the HV volatility graph for all ETFs under consideration is reported in Fig. 7.

Fig. 7 HV reported for all the TFT under examination

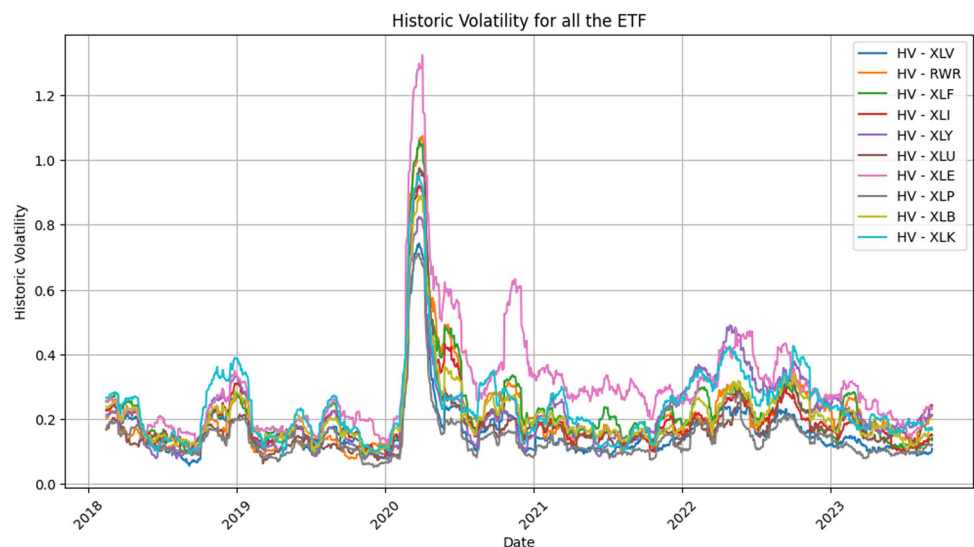


Table 11 MLP architecture hyperparameters

Layer Type	Hyperparameter Choices
Dense	Units: [32, 64, 128, 256] - Activation: ['relu', 'tanh']
Dropout	Rate: [0.25, 0.5]
Dense	Units: [32, 64, 128, 256] - Activation: ['relu', 'tanh']
Dropout	Rate: [0.25, 0.5]
Dense	Units: [8, 16, 32, 64] - Activation: ['relu', 'tanh'] - Regularization: L2
Dense	Units: [8, 16, 32, 64] - Activation: ['relu', 'tanh'] - Regularization: L2
Dropout	Rate: [0.25, 0.5]
Output (Dense)	Units: 1
Learning Rate	[1e-2, 1e-3, 1e-4]

Appendix B: Multilayer perceptron (MLP)

A multilayer perceptron (MLP) is a type of feedforward artificial neural network (ANN) characterized by its multiple layers, including an input layer, one or more hidden layers, and an output layer. This architecture enables MLPs to model complex relationships in data through the use of nonlinear activation functions. The key components of an MLP are:

- **Input Layer:** Receives the input signal for processing.
- **Hidden Layers:** Essential for computations and feature extraction, their number and the count of nodes within them are critical hyperparameters.
- **Output Layer:** Produces the final prediction or classification result.

Table 11 provides an in-depth look at the hyperparameters chosen for the MLP architecture.

Appendix C: Long short-term memory (LSTM)

Long short-term memory (LSTM) networks, a specialized type of recurrent neural network (RNN), are adept at addressing the vanishing gradient problem inherent in traditional RNNs. Their design makes them particularly suitable for tasks involving sequential data, such as time series forecasting, natural language processing, and sequence prediction. The architecture of an LSTM network comprises:

- **Input Gate:** Determines which values from the input are vital to update the memory state.
- **Forget Gate:** Decides which information should be removed from the cell state.
- **Output Gate:** Controls what the next hidden state should be, using it for future predictions.
- **Cell State:** Maintains relevant information throughout the sequence processing.

Table 12 provides an in-depth look at the hyperparameters chosen for the LSTM architecture.

Table 12 LSTM architecture hyperparameters

Layer Type	Hyperparameter Choices
LSTM (Input)	Units: [16, 32, 64, 128, 256] (return sequences: True)
LSTM	Units: [16, 32, 64, 128, 256] (return sequences: True)
LSTM	Units: [16, 32, 64, 128, 256] (return sequences: False)
Dense	Units: [16, 32, 64, 128] - Activation: ['relu', 'tanh'] - Regularization: L2
Dropout	Rate: [0.25, 0.5]
Dense	Units: [16, 32, 64, 128] - Activation: ['relu', 'tanh'] - Regularization: L2
Output (Dense)	Units: 1
Learning Rate	[1e-2, 1e-3, 1e-4]

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Data availability Data employed in this work can be obtained using the open-source tool of *Yahoo Finance* (<https://help.yahoo.com/kb/SLN2311.html>). In the present study, we covered the period from January 1, 2002, to October 31, 2023, collecting several features for each asset taken into account for this research (as explained in Section 3.1).

Declarations

Conflict of interest The authors declare no conflict of interest.

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